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Editorial Board
PLOS ONE

Dear Editors,

We submit for your consideration the manuscript entitled “Code Geometry and Endogenous Enforcement Concentration in an Agent-Based Model of Fundamentalism.”

The manuscript presents an agent-based model demonstrating that enforcement-dominant dynamics in codified moral systems (the behavioral syndrome typically labeled “fundamentalism”) arise endogenously from measurable structural properties of the code rather than from theological content, abnormal psychology, or exogenous political shocks. The model identifies five institutional dimensions—legibility, substitutability, enforcement reward, institutional delegation, and exit cost—whose joint configuration determines which of four enforcement regimes a system enters: quiet, mixed, collapse, or capture.

Three features distinguish this work from existing computational and sociological treatments.

First, a fine-grained dose-response sweep (1,200 runs across ten legibility values) identifies a sharp activation threshold in code legibility below which enforcement cannot engage regardless of other parameters. This threshold corresponds to a structural feature of how codified systems relate truth to persons: systems with transpersonal authority architectures (where compliance is internal and unmonitorable) fall below the threshold; systems with person-dependent authority architectures (where compliance is behavioral and externally auditable) fall above it.

Second, ablation experiments (180 matched runs) confirm that enforcement concentration is genuinely endogenous. Disabling the institutional monopoly restriction leaves concentration metrics unchanged, ruling out the possibility that the result is mechanically imposed by the model architecture.

Third, the model demonstrates that endogenous narrative drift—the enforcement apparatus’s degradation of perceived outside options—is the specific mechanism converting enforcement concentration into durable capture. Two theoretically motivated alternatives (sanctified suffering and membership reward) are empirically inert.

The manuscript reports approximately 5,000 simulation runs across eight experimental specifications, including the baseline sweep, confirmatory analysis, retention mechanism sweeps, endogenous drift sweep, monopoly ablation, cadre quota sensitivity, intensity gate sensitivity, and the dose-response sweep. All source code, parameter configurations, analysis scripts, and supporting data will be deposited in a public repository with a persistent DOI prior to publication.

A companion paper analyzing the philosophical foundations of transpersonal authority architecture in the Vedic tradition and its institutional consequences across six religious comparison cases has been submitted separately to the *Journal of Hindu Studies* (Boggavarapu, “*Apaurusheya* as Struc-

tural Epistemology”). The two papers are complementary but independent: the present manuscript stands on its computational results without requiring the philosophical analysis, and the philosophical paper does not depend on the ABM.

The manuscript has not been submitted elsewhere and is not under consideration at another journal. There are no competing interests to declare.

We believe this work is appropriate for PLOS ONE because it presents a technically rigorous computational model with falsifiable predictions, transparent methodology, and publicly available code and data, evaluated against the journal’s criteria of methodological soundness and reproducibility.

Sincerely,

[Author Name]

Affiliation